



SERANGOON JUNIOR COLLEGE
General Certificate of Education Advanced Level
Higher 1

Candidate Name

Class

CHEMISTRY

Preliminary Examination
Paper 1 Multiple Choice

8872/01

30 August 2013
1 hour

Additional Materials: Data Booklet
 Optical Mark Sheet (OMS)

READ THESE INSTRUCTIONS FIRST

On the separate multiple choice OMS given, write your name, FIN/NRIC and class in the spaces provided.

Shade correctly your class and FIN/NRIC number.

Eg. If your NRIC is S9306660Z, shade **S9306660Z** for the item "index number".

There are **thirty** questions in this paper. Answer **all** questions.

For each question there are four possible answers **A, B, C** and **D**.

Choose the one you consider correct and record your choice using a **soft pencil** on the separate OMS.

Each correct answer will score one mark.

A mark will not be deducted for a wrong answer.

You are advised to fill in the OMS as you go along; no additional time will be given for the transfer of answers once the examination has ended.

Any rough working should be done in this question paper.

Section A
Answer all questions

- 1 The percentage by mass of water in a hydrated manganese(II) chloride salt is 36.4%. What is the empirical formula of the hydrated salt?
- A $\text{MnCl}_2 \cdot 2\text{H}_2\text{O}$
- B $\text{MnCl}_2 \cdot 3\text{H}_2\text{O}$
- C $\text{MnCl}_2 \cdot 4\text{H}_2\text{O}$
- D $\text{MnCl}_2 \cdot 5\text{H}_2\text{O}$
- 2 In which one of the following reactions is there a **reduction** in the oxidation state of the underlined element?
- A $\underline{\text{N}}\text{H}_3 + \text{HCl} \rightarrow \underline{\text{N}}\text{H}_4\text{Cl}$
- B $2\underline{\text{V}}^{3+} + \text{H}_2\text{O}_2 \rightarrow 2\underline{\text{V}}\text{O}^{2+} + 2\text{H}^+$
- C $\text{H}_2\underline{\text{O}}_2 + 2\text{I}^- + 2\text{H}^+ \rightarrow 2\text{H}_2\underline{\text{O}} + \text{I}_2$
- D $\text{Ca}\underline{\text{C}}\text{O}_3 \rightarrow \text{CaO} + \underline{\text{C}}\text{O}_2$
- 3 Which of the following particles has more electrons than protons, and more protons than neutrons? [$\text{H} = {}^1_1\text{H}$, $\text{D} = {}^2_1\text{H}$, $\text{He} = {}^4_2\text{He}$, $\text{O} = {}^{16}_8\text{O}$]
- A He^+
- B D^-
- C OD^-
- D OH^-

- 4 Under the same conditions, which of the following ions would be deflected in a mass spectrometer to the same extent as $^{12}_6\text{C}^{2+}$?
- (i) $^6_3\text{Li}^+$
 - (ii) $^{12}_6\text{C}^+$
 - (iii) $^{12}_7\text{N}^{2+}$
 - (iv) $^{13}_6\text{C}^{2+}$
- A (i), (ii), (iii) and (iv)
- B (i) and (iii) only
- C (ii) only
- D None of the above.
- 5 Which of the following statements best explains why the boiling point of $\text{CH}_3\text{CH}_2\text{CH}_2\text{NH}_2$ is higher than that of $(\text{CH}_3)_3\text{N}$?
- A $\text{CH}_3\text{CH}_2\text{CH}_2\text{NH}_2$ has a higher relative molecular mass than $(\text{CH}_3)_3\text{N}$.
- B The $\text{CH}_3\text{CH}_2\text{CH}_2\text{NH}_2$ molecule has a larger surface area than the $(\text{CH}_3)_3\text{N}$ molecule.
- C $\text{CH}_3\text{CH}_2\text{CH}_2\text{NH}_2$ is polar but $(\text{CH}_3)_3\text{N}$ is not.
- D There are hydrogen bonds between $\text{CH}_3\text{CH}_2\text{CH}_2\text{NH}_2$ molecules but not between $(\text{CH}_3)_3\text{N}$ molecules.
- 6 Although F–F bond is the shortest among the halogens, it has the weakest bond strength. What could be the possible reason for the unusual phenomenon?
- A Fluorine has the highest electronegativity in Group VII.
- B Fluorine has the smallest atomic radius in Group VII.
- C Due to lone pair-lone pair repulsion between the 2 fluorine atoms.
- D Due to the non-polar nature of the F–F bond.

- 7 Orange dichromate, $\text{Cr}_2\text{O}_7^{2-}$ and yellow chromate, CrO_4^{2-} exist in equilibrium in aqueous solution.

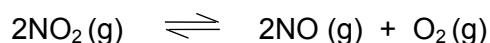


Which statement about this equilibrium is correct?

- A $\text{Cr}_2\text{O}_7^{2-}$ is reduced to CrO_4^{2-} .
B The solution appears to be yellow in NaOH (aq).
C K_c value increases with the addition of NaOH (aq).
D Lowering of pH will increase the concentration of CrO_4^{2-} .
- 8 Each of the following equilibria is subjected to two changes which are carried out separately:
(i) The pressure is reduced at constant temperature;
(ii) The temperature is increased at constant pressure.

For which equilibrium will **both** of these changes result in an increase in the proportion of products?

- A $\text{H}_2(\text{g}) + \text{I}_2(\text{g}) \rightleftharpoons 2\text{HI}(\text{g}); \quad \Delta H = +53 \text{ kJ mol}^{-1}$
B $4\text{NH}_3(\text{g}) + 5\text{O}_2(\text{g}) \rightleftharpoons 4\text{NO}(\text{g}) + 6\text{H}_2\text{O}(\text{g}); \quad \Delta H = -950 \text{ kJ mol}^{-1}$
C $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g}); \quad \Delta H = -92 \text{ kJ mol}^{-1}$
D $\text{N}_2\text{O}_4(\text{g}) \rightleftharpoons 2\text{NO}(\text{g}); \quad \Delta H = +57 \text{ kJ mol}^{-1}$
- 9 Nitrogen dioxide decomposes on heating according to the following equation



When 4 moles of nitrogen dioxide was put into a 1 dm^3 container and heated, the equilibrium mixture contained 0.8 moles of oxygen. What is the numerical value of the equilibrium constant, K_c , at the temperature of the experiment?

- A $\frac{1.6 \times 0.8}{2.4^2}$
B $\frac{1.6^2 \times 0.8}{2.4^2}$
C $\frac{0.8 \times 0.8}{2.4}$
D $\frac{0.8^2 \times 0.8}{4^2}$

- 10** Values for the ionic product of water, K_w , at two different temperatures are given below.

Temperature / °C	$K_w / \text{mol}^2 \text{dm}^{-6}$
25	1.00×10^{-14}
30	1.44×10^{-14}

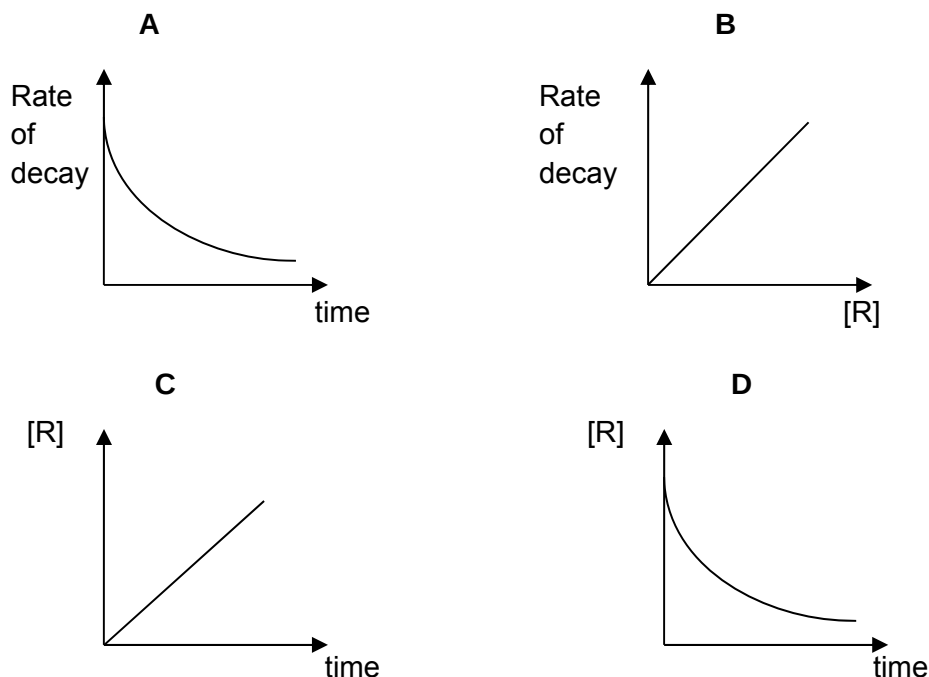
Which is correct for pure water at 30 °C?

- A** pH < 7
 - B** $[\text{H}^+] < [\text{OH}^-]$
 - C** $[\text{H}^+] = 1.44 \times 10^{-7} \text{ mol dm}^{-3}$
 - D** Dissociation of water is an exothermic process
- 11** Lead is the final product formed by a series of changes in which the rate-determining stage is the radioactive decay of uranium-238. This decay is a first-order reaction with half-life of 4.5×10^9 years.

What would be the age of a rock sample originally lead-free, in which molar ratio of uranium to lead is now 0.143?

- A** 1.5×10^9 years
- B** 4.5×10^9 years
- C** 12.6×10^9 years
- D** 13.5×10^9 years

- 12 A reaction involving reactant **R** is found out to be first order. Which of the following graphs best describes the decay of reactant **R**?



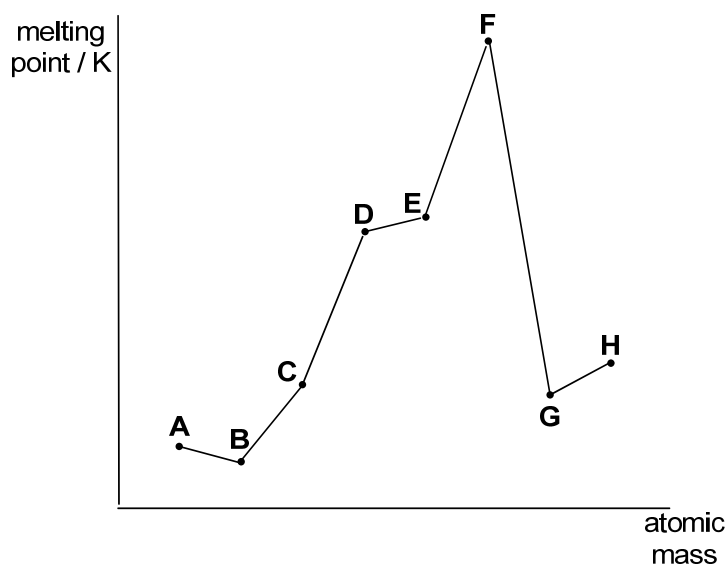
- 13 The magnitude of enthalpy change of neutralisation of aqueous sodium hydroxide by hydrochloric acid is 57.2 kJ mol^{-1} , but the magnitude of enthalpy change of neutralisation of aqueous sodium hydroxide by aqueous ethanoic acid is 55.2 kJ mol^{-1} .

Which statement best explains the numerical difference in these values?

- A The process $\text{CH}_3\text{CO}_2\text{H (aq)} \rightarrow \text{CH}_3\text{CO}_2^- \text{ (aq)} + \text{H}^+ \text{ (aq)}$ is exothermic
- B The process $\text{CH}_3\text{CO}_2\text{H (aq)} \rightarrow \text{CH}_3\text{CO}_2^- \text{ (aq)} + \text{H}^+ \text{ (aq)}$ is endothermic
- C Aqueous ethanoic acid is a weak acid and has fewer hydrogen ions than hydrochloric acid of the same concentration.
- D A smaller volume of aqueous hydrochloric acid is required for the neutralisation compared to that of ethanoic acid of the same concentration.

- 14 SiCl_4 is known to hydrolyse readily in water but CCl_4 is inert to water. Which statement best explains this observation?
- A CCl_4 has a giant molecular structure, large amount of energy is required to overcome the strong covalent bond to allow solvation.
 - B There are vacant orbitals to accommodate additional electrons in SiCl_4 but not in CCl_4 .
 - C CCl_4 has weaker van der Waals' forces of attraction which cannot displace the stronger hydrogen bonding in water.
 - D Si^{4+} has high charge density to polarise the large chlorine electron cloud.
- 15 *Use of the Data Booklet is relevant to this question.*

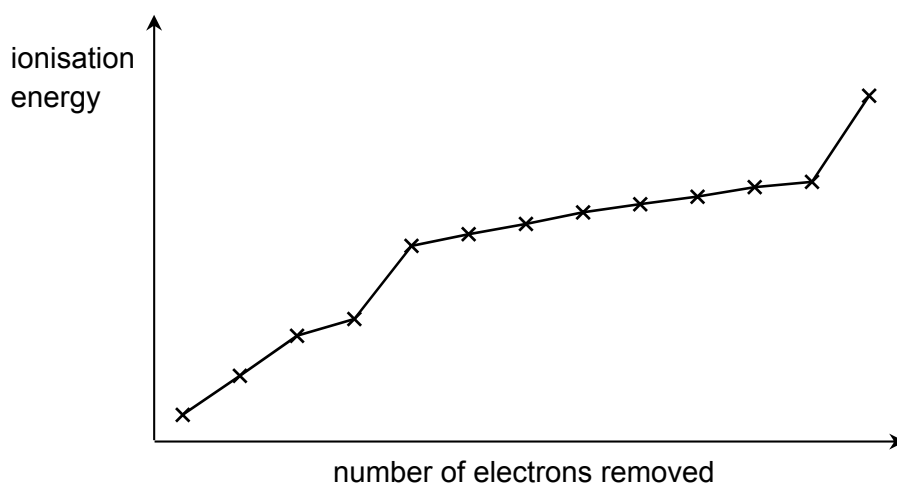
The graph below shows the variation in the melting points for eight consecutive elements in the Periodic Table, all with atomic number below 20.



What statement is correct?

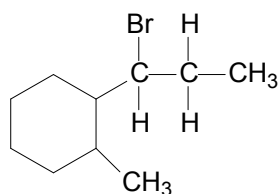
- A Element F conducts electricity at room temperature.
- B Element D burns with a brilliant yellow flame.
- C Element H does not react with air at room temperature.
- D Element C is a gas which is chemically inert at room temperature.

- 16 The graph below shows the first thirteen ionisation energies for element J.



What can be deduced from the graph about element J?

- A It is a d-block element.
 B It is in Group IV of the Periodic Table.
 C It has one electron in its outermost shell.
 D It is in the third period (Na to Ar) of the Periodic Table.
- 17 How many different alkenes, including geometrical isomers, could be produced when



reacts with hot ethanolic NaOH?

- A 2 B 3 C 4 D 5
- 18 2,2-dimethylbutane undergoes free radical substitution with chlorine under ultraviolet light. How many different mono-substituted chlorinated products are possible?

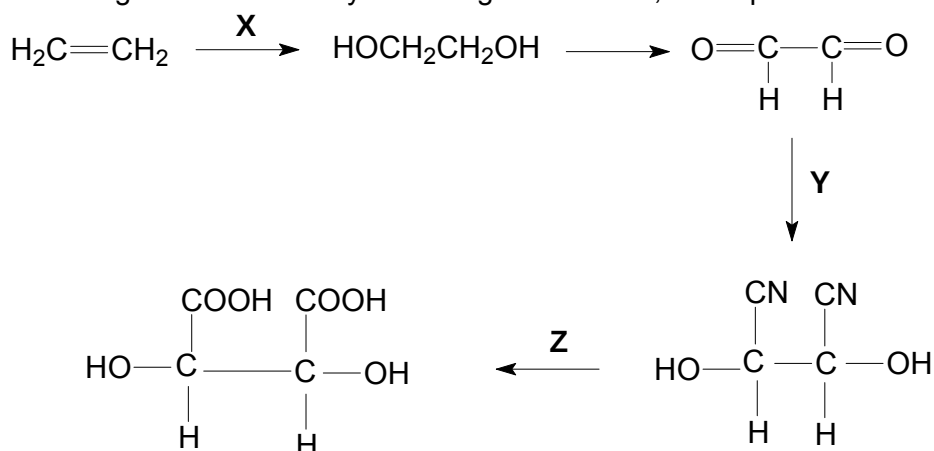
- A 2 B 3 C 4 D 5

- 19 Chlorofluorocarbons (CFCs) have been widely used in aerosol sprays, refrigerators and in making foamed plastics, but are now known to destroy ozone in the upper atmosphere.

Which of the following will not destroy ozone, and therefore can be used as a safer replacement for CFCs?

- A $\text{CH}_2\text{FCH}_2\text{CH}_2\text{CH}_3$
- B CCl_3CBr_3
- C CHBr_3
- D CHClFCClF_2

- 20 The following is a method of synthesising tartaric acid, a compound found in wine.



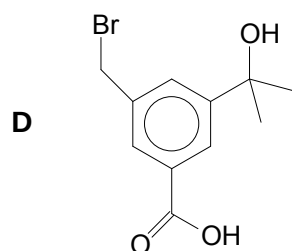
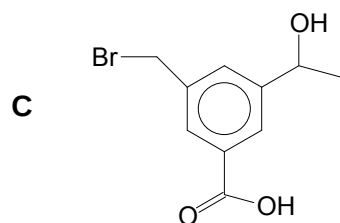
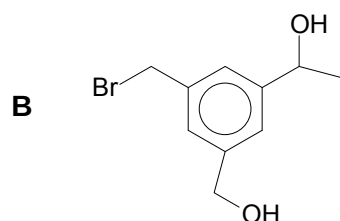
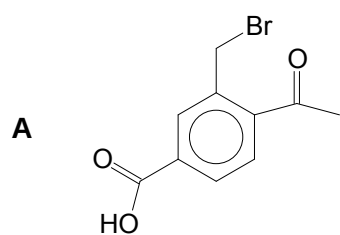
Which set of reagents and conditions can be used for the synthesis?

- | | Stage X | Stage Y | Stage Z |
|---|--|--|--|
| A | $\text{H}_2\text{O} (\text{g}), \text{H}_3\text{PO}_4 (\text{aq})$ | Cold $\text{HCN}, \text{NaOH} (\text{aq})$ | Hot $\text{K}_2\text{Cr}_2\text{O}_7, \text{H}_2\text{SO}_4 (\text{aq})$ |
| B | Cold $\text{KMnO}_4, \text{H}_2\text{SO}_4 (\text{aq})$ | Ethanolic KCN , heat | $\text{HCl} (\text{aq})$, heat |
| C | $\text{H}_2\text{O} (\text{g}), \text{H}_3\text{PO}_4 (\text{aq})$ | Ethanolic KCN , heat | Hot $\text{K}_2\text{Cr}_2\text{O}_7, \text{H}_2\text{SO}_4 (\text{aq})$ |
| D | Cold $\text{KMnO}_4, \text{NaOH} (\text{aq})$ | Cold $\text{HCN}, \text{NaOH} (\text{aq})$ | $\text{HCl} (\text{aq})$, heat |

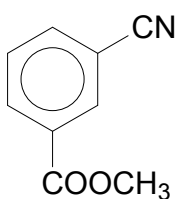
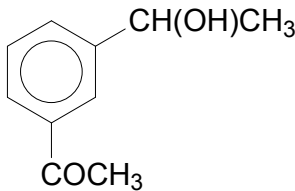
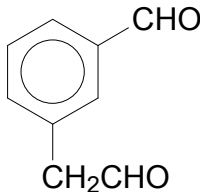
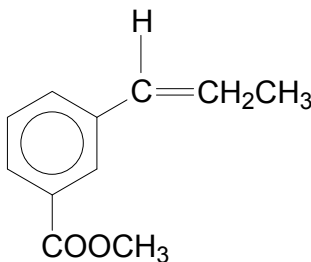
21 Data about the reactions of compound **K** are given in the table.

Reaction	Reagent	Result
1	Hot acidified potassium dichromate, followed by 2,4-DNPH	Orange solution turns green and orange precipitate seen after addition of 2,4-DNPH.
2	Warm alkaline aqueous Iodine	Yellow precipitate observed.
3	Aqueous sodium carbonate	Effervescence seen.
4	AgNO ₃ (aq), warm	Cream solid formed.

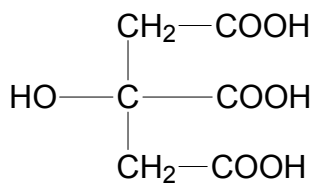
Which of the following compounds is **K**?



22 Which of the following reactions will **not** produce 1,3-dibenzoic acid?

- A**  + aqueous H_2SO_4 , reflux
- B**  + alkaline aqueous iodine followed by acidification
- C**  + Tollens' reagent followed by acidification
- D**  + hot acidified KMnO_4

23 Citric acid can be found in lemon juice and orange juice and it causes the sharp unique taste.



Which of the following reacts completely with 1 mole of citric acid?

- A** 3 moles of PCl_5
- B** 3 moles of NaHCO_3
- C** 4 moles of H_2SO_4
- D** 4 moles of KOH

24 Arrange the following molecules in terms of increasing pK_a values.

Compound **L** $\text{CH}_3\text{CH}_2\text{CH}_2\text{COOH}$

Compound **M** $\text{CH}_2(\text{Br})\text{CH}_2\text{CH}_2\text{COOH}$

Compound **N** $\text{CH}_3\text{CH}_2\text{CH}(\text{Br})\text{COOH}$

Compound **P** $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{COOH}$

A $\text{M} < \text{N} < \text{L} < \text{P}$

B $\text{M} < \text{N} < \text{P} < \text{L}$

C $\text{N} < \text{M} < \text{L} < \text{P}$

D $\text{N} < \text{P} < \text{L} < \text{M}$

25 An ester, $\text{C}_4\text{H}_8\text{O}_2$, undergoes the following reactions:

- Reacts with hot sulfuric acid to give carboxylic acid **Q** and alcohol **R**.
- The product from the oxidation of alcohol **R** gives a brick red precipitate with hot alkaline copper (II) complex.
- Carboxylic acid **Q** reacts with lithium aluminium hydride in dry ether to give **R**.

Which of the following could be the structure of the ester?

A $\text{HCOOCH}(\text{CH}_3)_2$

B $\text{CH}_3\text{COOCH}_2\text{CH}_3$

C $\text{HCOOCH}_2\text{CH}_2\text{CH}_3$

D $\text{CH}_3\text{OCOCH}_2\text{CH}_3$

Section B

For **questions 26-30**, one or more of the numbered statements **1** to **3** may be correct. Decide whether each of the statements is or is not correct. The responses **A** to **D** should be selected on the basis of:

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

No other combination of statements is to be used as a correct response.

26 Which one of the following statements is **true**?

- 1 AlCl_3 is covalent while AlF_3 is ionic.
- 2 The melting point of Na is lower than that of Mg.
- 3 An aqueous solution of HBr conducts electricity because HBr is ionic.

27 Which of the following mixtures will result in an acidic buffer solution?

- 1 25 cm^3 of $\text{CH}_3\text{COO}^-\text{Na}^+$ and 25 cm^3 of CH_3COOH of same concentration
- 2 25 cm^3 of $\text{CH}_3\text{COO}^-\text{Na}^+$ and 12.5 cm^3 of HCl of same concentration
- 3 25 cm^3 of NaOH and 12.5 cm^3 of CH_3COOH of same concentration

28 Why is a solution of aluminium chloride more acidic than magnesium chloride?

- 1 The O-H bonds are weaker in $[\text{Al}(\text{H}_2\text{O})_6]^{3+}$ than in H_2O .
- 2 Chloride ions react with water to form hydrochloric acid.
- 3 Aluminium atom has high charge density.

29 Which of the following statements are true about the chemical reactions of cyclohexane, cyclohexene and methylbenzene?

- 1 One mole each of cyclohexene and methylbenzene will react with one mole and three moles of hydrogen gas respectively when heated in the presence of a catalyst under appropriate condition.
- 2 All three compounds decolourise bromine gas under *uv* light.
- 3 All three compounds decolourise hot acidified manganate (VII).

- 30** Which starting materials will give sodium benzoate on heating with aqueous sodium hydroxide under reflux?
- 1** phenyl ethanoate
 - 2** phenyl benzoate
 - 3** ethylbenzene and sodium manganate (VII)

END